Bivariate & Multivariate Visualization

Wook-Shin Han

November 10, 2025

Overview — Plot Families & What We'll Use

Relational (X–Y)

- Scatter (`scatterplot`)
- Line (`lineplot`)
- Regression / LOESS (`regplot`, `lmplot`)
- Figure-level faceting (`relplot`)

Bivariate Density

- Hexbin / 2D Hist (`jointplot(kind="hex"/"hist")`)
- 2D KDE (`jointplot(kind="kde")`)

Distributions (by group)

- Hist/KDE (`displot`, `kdeplot`)
- ECDF (`ecdfplot`)
- Box / Violin / Boxen (`boxplot`, `violinplot`, `boxenplot`)
- Raw points (`stripplot`/`swarmplot`)

Matrix / Structure

- Pair plot (`pairplot`)
- Correlation heatmap (`heatmap`)
- Clustered heatmap (`clustermap`)

Multivariate Projections

- PCA scatter (`PCA` + `scatterplot`)
- Parallel coordinates

Diagnostics

- Residuals vs Fitted

Faceting & Encodings

- `hue`, `style`, `size`, `row`/`col`, `col_wrap`

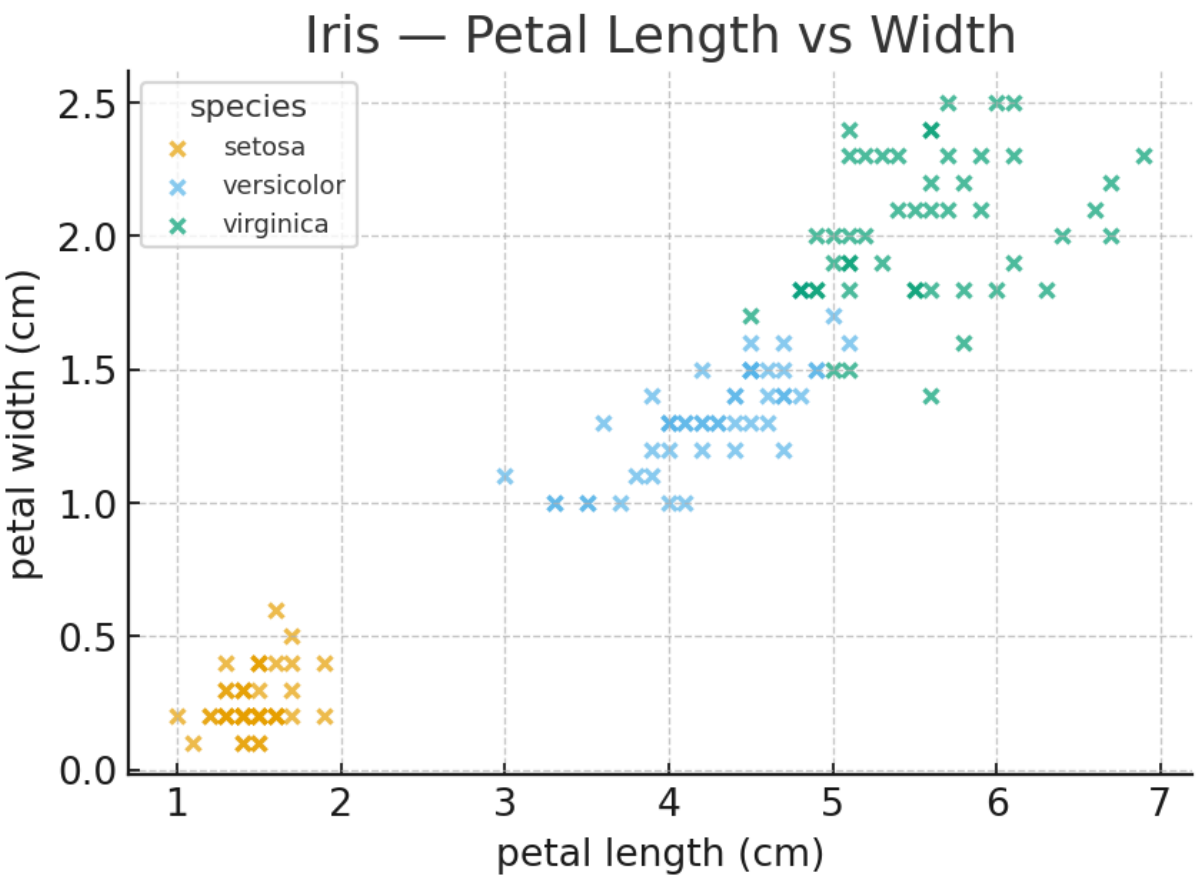
Datasets: Iris, Wine, Diabetes, Synthetic (`df_long`).

Scatter — Definition & When to Use

Definition. Points at (x_i, y_i); optional encodings: *hue* (category), *size* (numeric), *style* (marker).
Use when. Exploring linear/nonlinear relations, clusters, outliers.
How to read. Slope and shape; band thickness (variance); separate clouds $\Rightarrow$ clusters.
Pitfalls. Overplotting at large n; transparency or density plots help.

Scatter — Code & Executed Example

```
sns.scatterplot(data=iris,
                x="petal length (cm)", y="petal width (cm)",
                hue="species", alpha=0.7)
```



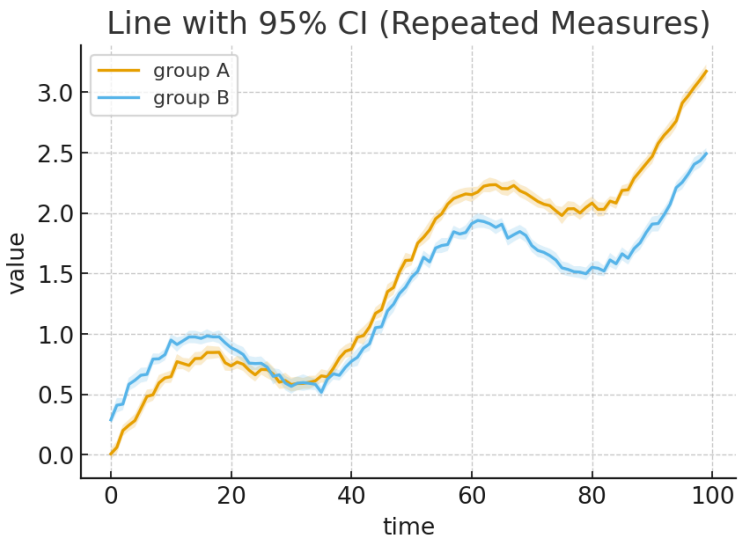
Line — Definition & When to Use

Definition. Connect observations along ordered x (time, dose).
Use when. Trends over time, evolution curves, means with uncertainty bands.
How to read. Slope (rate of change), curvature (acceleration), CI band width (stability).
Pitfalls. Do not connect unordered categories; watch for unequal sampling intervals.

Line — Data Generation & Code (with CI)

```python
# Build df_long with repeated measures per time & group
rng = np.random.default_rng(42)
T, n_rep = 100, 30
time = np.arange(T); rows = []
for g, base, slope in [("A",0.0,0.03),("B",0.3,0.02)]:
    for r in range(n_rep):
        y = base + slope*time + 0.4*np.sin(time/7.5) + rng.normal(0,0.15,T)
        rows.append(pd.DataFrame({"time":time,"value":y,"group":g,"rep":r})
            )
df_long = pd.concat(rows, ignore_index=True)

sns.lineplot(data=df_long, x="time", y="value",
             hue="group", errorbar=("ci", 95))
```

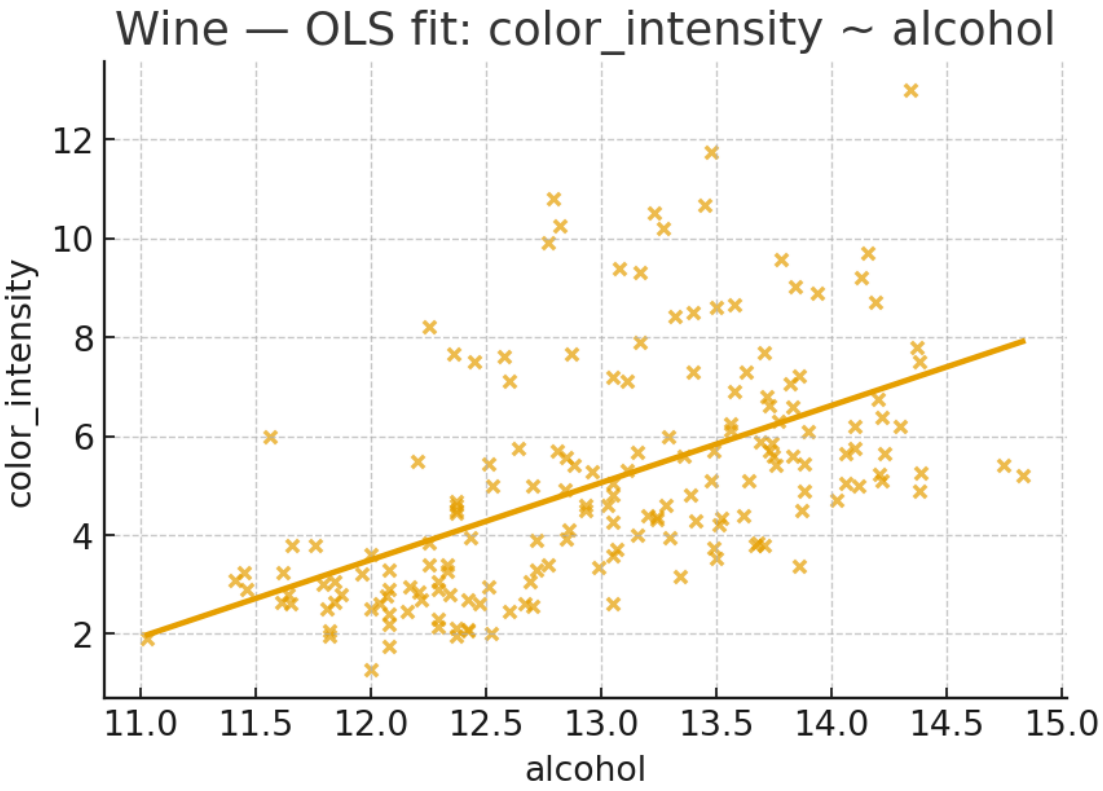


Regression (OLS/LOESS) — Definition & When to Use

Definition. OLS fits $E[Y|X] = \beta_0 + \beta_1 X$; LOESS fits a locally weighted smoother.
Use when. Summarizing central trend; comparing slopes by group.
How to read. Slope sign/magnitude; $\pm$ CI envelope; residual diagnostics for fit quality.
Pitfalls. OLS sensitive to outliers; LOESS bandwidth controls smoothness.

Regression — Code & Executed Example

```
sns.regplot(data=wine, x="alcohol", y="color_intensity")
# Local smoother
sns.lmplot(data=wine, x="alcohol", y="color_intensity", lowess=
    True)
```

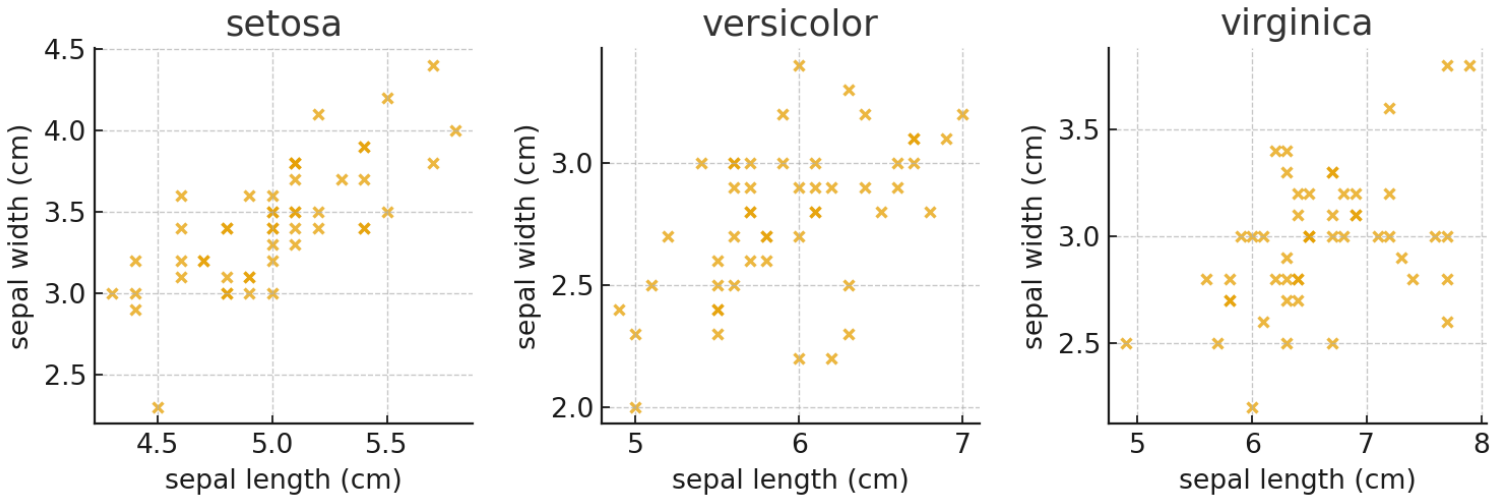


relplot — Definition & When to Use

Definition. Figure-level wrapper for `scatterplot`/`lineplot` with built-in faceting (`row`, `col`), legends, and consistent sizing.
Use when. You want small multiples of relationships by subgroup(s) with one call (`kind="scatter"` or `"line"`).
How to read. Compare shapes/slopes across facets with shared axes; `hue`/`style`/`size` add encodings.
Pitfalls. Too many facets become unreadable; prefer `col_wrap` and keep encodings minimal.

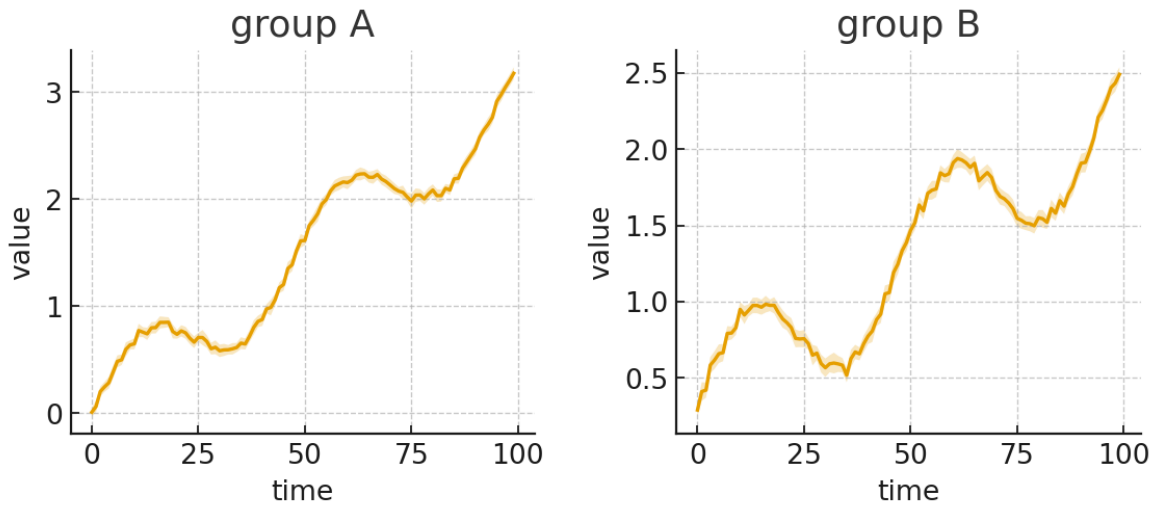
relplot — Scatter with Facets (Iris)

```
sns.relplot(
    data=iris,
    x="sepal length (cm)", y="sepal width (cm)",
    hue="species", col="species", kind="scatter",
    col_wrap=3, height=3.0, aspect=1.05, alpha=0.75
)
```



relplot — Line with CI & Facets (Repeated Measures)

```
sns.relplot(
    data=df_long, x="time", y="value",
    hue="group", col="group", col_wrap=2,
    kind="line", errorbar=("ci",95),
    height=3.0, aspect=1.25
)
```

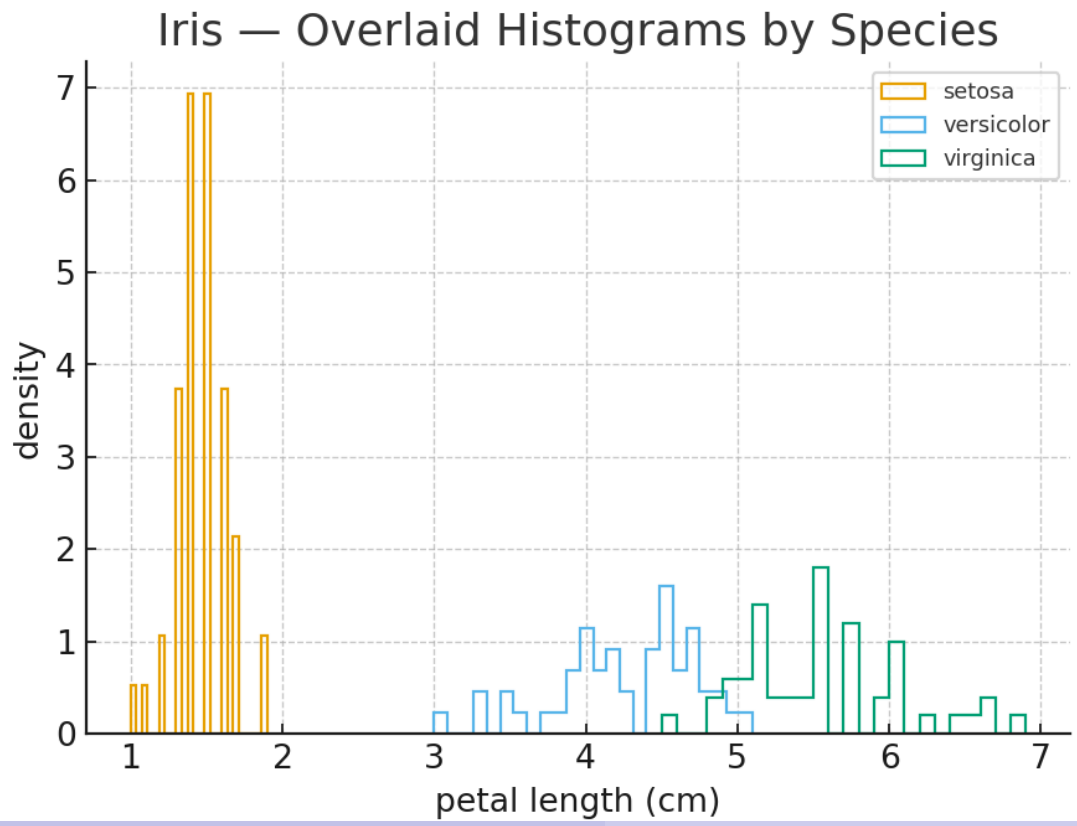


Multiple Distributions — Definition & When to Use

Definition. Compare distributions across groups with figure-level `displot` (hist/KDE) or axes-level `kdeplot`/`histplot`/`ecdfplot`.
Use when. You need fair, repeatable comparisons of shapes/locations across categories.
How to read. Location (shift), spread, skew/tails, multimodality; stacked/fill show composition.
Pitfalls. Normalize carefully: `stat="density"` & `common_norm=False` avoid misleading areas across groups.

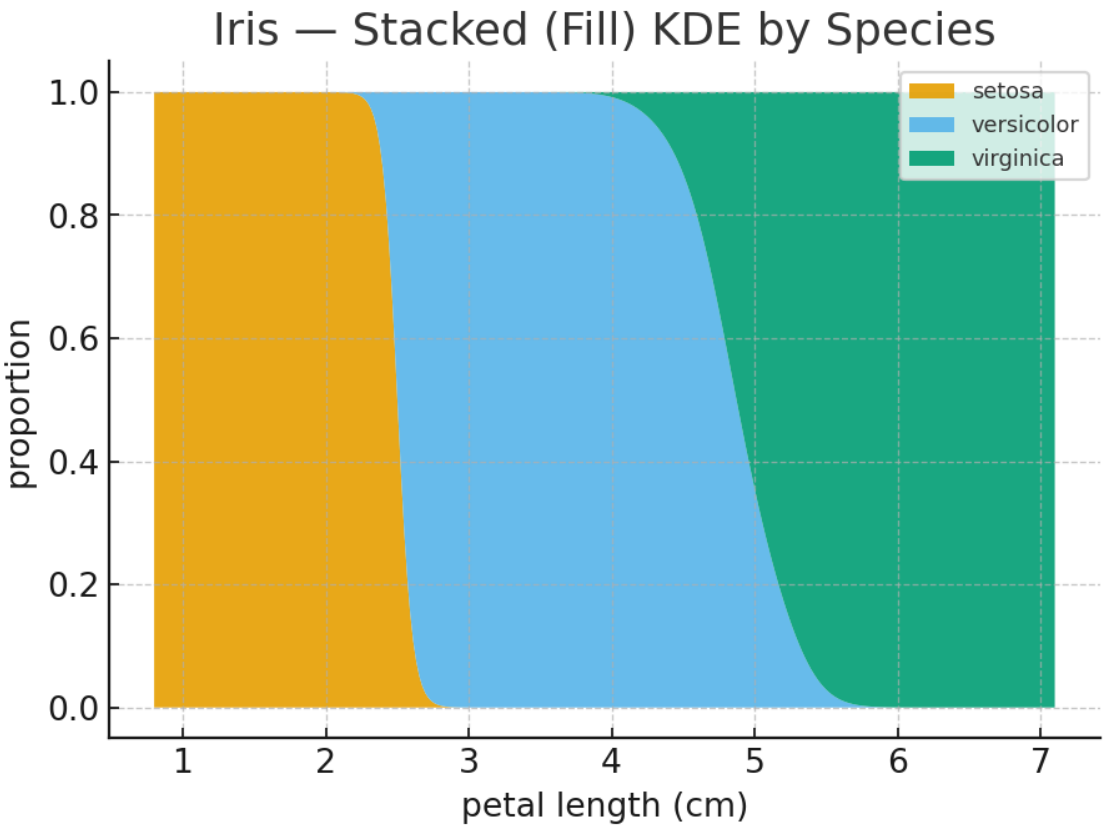
displot — Overlaid Histograms by Group (Iris)

```
sns.displot(
    data=iris,
    x="petal length (cm)", hue="species",
    kind="hist", bins=24,
    stat="density", common_norm=False,
    element="step"  # outlines for clean overlays
)
```



kdeplot — Stacked/Fill Multiple KDEs

```
sns.kdeplot(
    data=iris, x="petal length (cm)",
    hue="species", multiple="fill",
    common_norm=False, bw_adjust=0.9, alpha=0.9
)
```

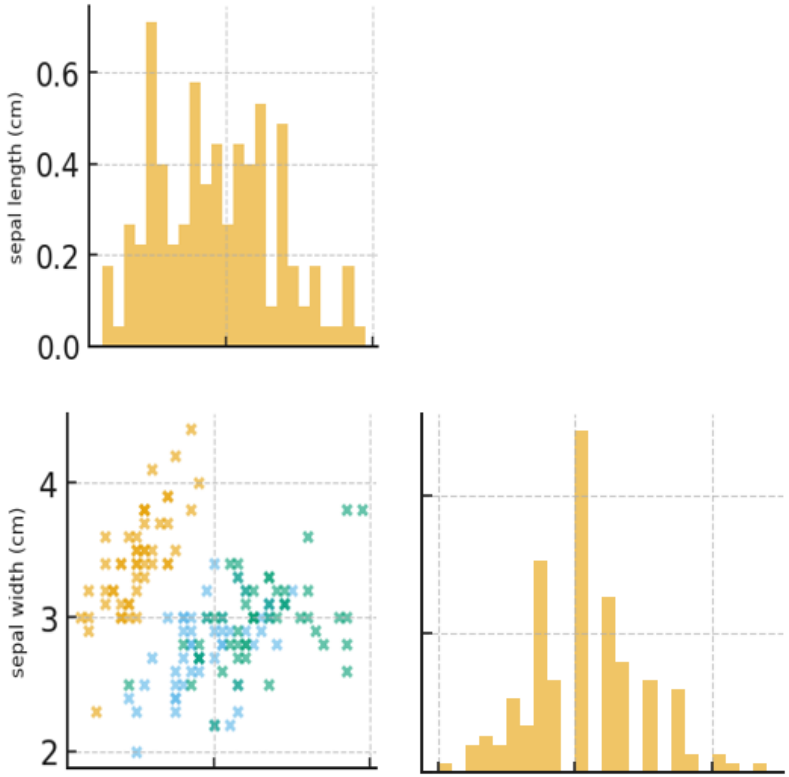


Pair Plot (Scatter-Matrix) — Definition & When to Use

Definition. Grid of pairwise scatter plots with univariate diagonals (`hist`/`kde`); a matrix of relations.
Use when. Quick scan of pairwise associations, clusters, separability by label.
How to read. Off-diagonals show relationships; diagonals show marginal shapes; `corner=True` removes duplicates.
Pitfalls. $O(d^2)$ axes: downsample for large n or many features.

Pair Plot — Code & Executed Example (Iris)

```
sns.pairplot(
  iris,
  vars=["sepal length (cm)","sepal width (cm)",
        "petal length (cm)","petal width (cm)"],
  hue="species", corner=True, diag_kind="kde",
  plot_kws=dict(alpha=0.6, s=22)
)
```

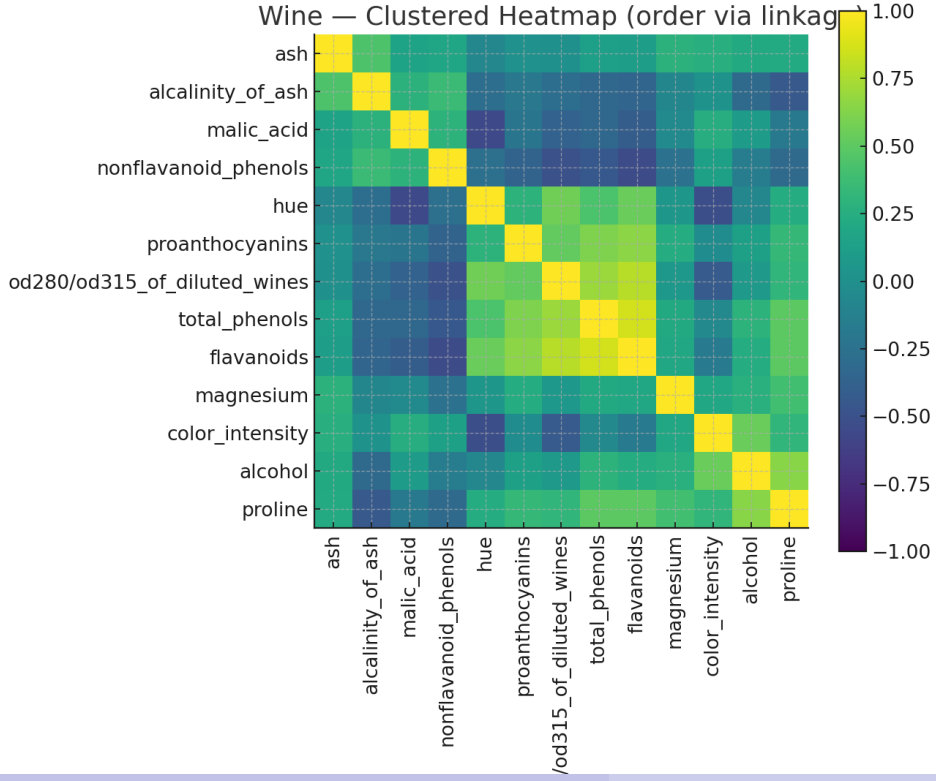


Clustermap — Definition & When to Use

Definition. Hierarchically clustered heatmap (rows/cols); reorders to reveal blocked structure.
Use when. Finding feature groups or sample clusters; exploring correlation blocks.
How to read. Dendrograms indicate similarity; blocks on the diagonal suggest modules.
Pitfalls. Sensitive to distance/linkage; standardize when mixing scales; avoid too many labels.

Clustermap — Code & Executed Example (Wine Correlations)

```
num = wine.drop(columns=["class","target"], errors="ignore")
corr = num.corr()
sns.clustermap(
    corr, center=0, cmap="vlag",
    linewidths=0.3, method="average", metric="euclidean"
)
```

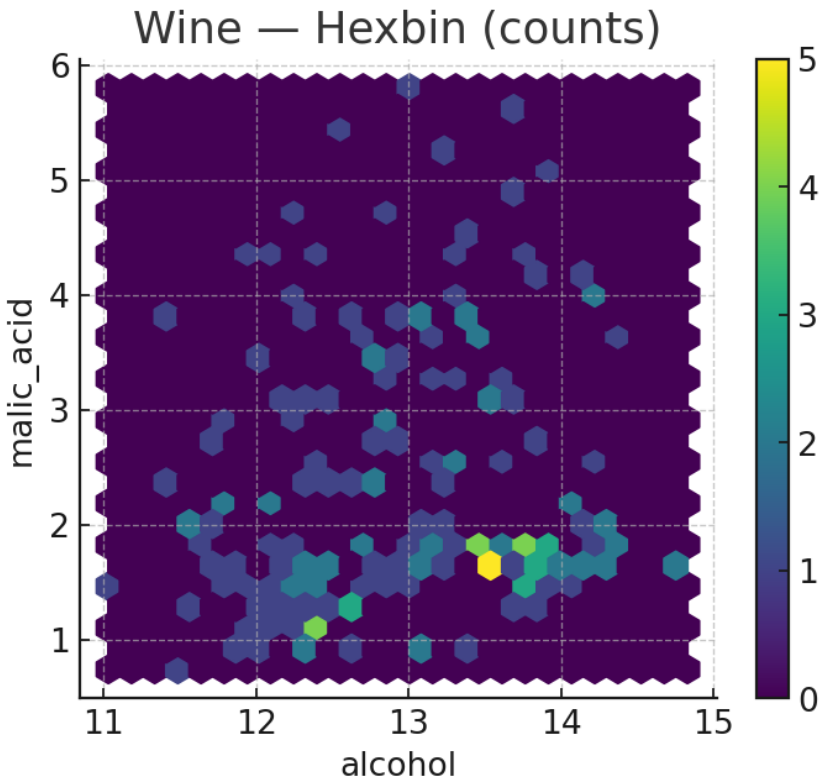


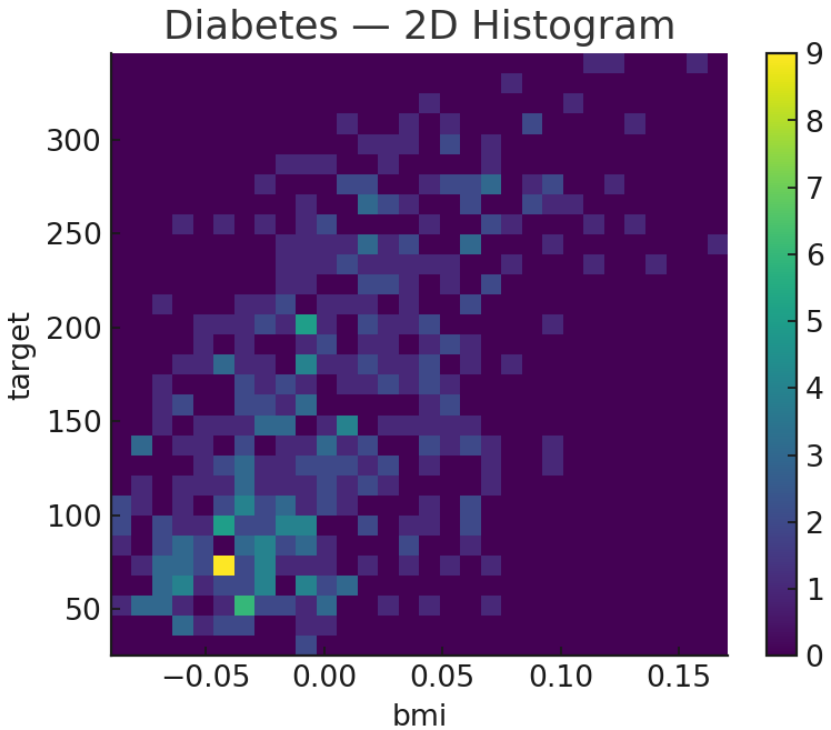
Hexbin & 2D Histogram — Definition & When to Use

Definition. Partition (x, y) domain into hexagonal (hexbin) or rectangular (2D hist) bins; color encodes counts.
Use when. Dense scatter clouds; need density view.
How to read. Hotspots = frequent combinations; band shape = relation form.
Pitfalls. Bin size affects perception; compare with multiple grids.

Hexbin / 2D Hist — Code & Executed Examples

```
sns.jointplot(data=wine, x="alcohol", y="malic_acid", kind="hex")
sns.jointplot(data=diabetes, x="bmi", y="target", kind="hist")
```



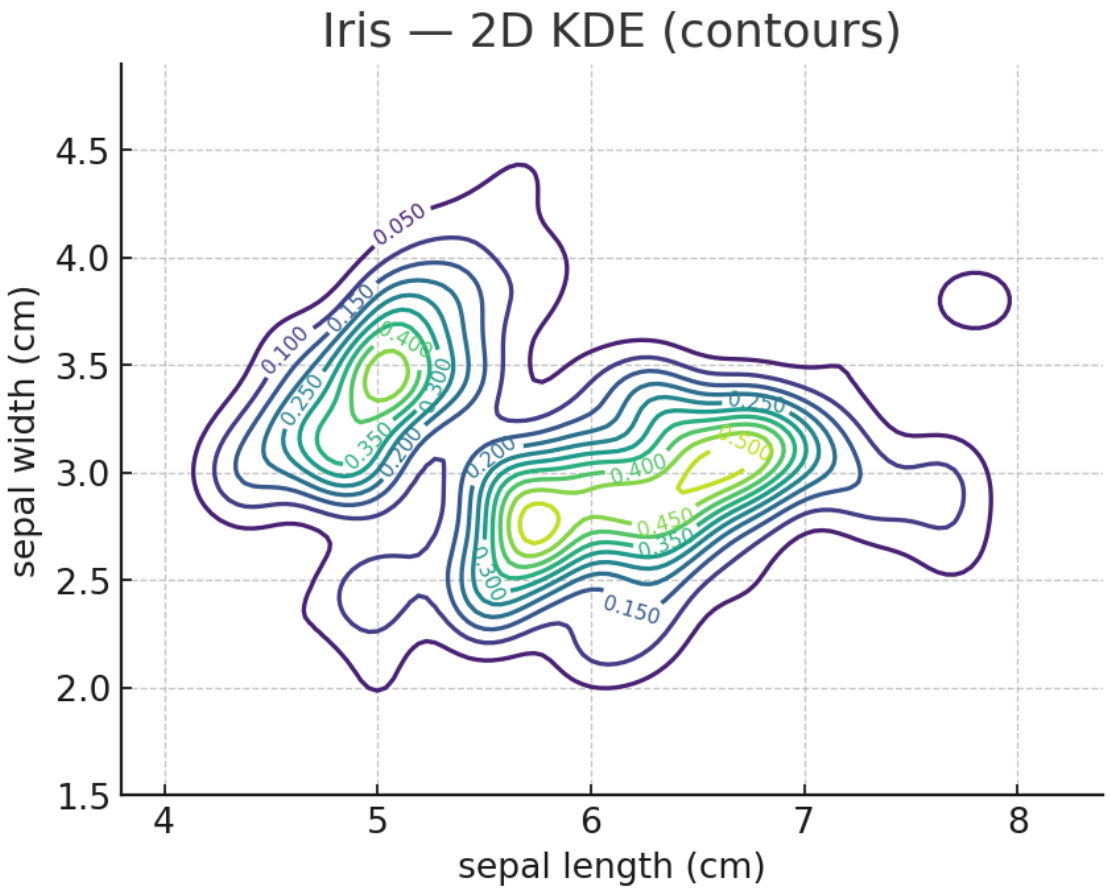


2D KDE — Definition & When to Use

Definition. Kernel density estimate in 2D: $\hat{f}(x, y) = \frac{1}{nh_x h_y} \sum K_x(\cdot) K_y(\cdot)$.
Use when. Smooth density structure, modes, ridges.
How to read. Contour peaks = high density; spacing reflects gradient.
Pitfalls. Bandwidth critical; use `bw_adjust`.

2D KDE — Code & Executed Example

```
sns.jointplot(data=iris,
              x="sepal length (cm)",
              y="sepal width (cm)", kind="kde")
```

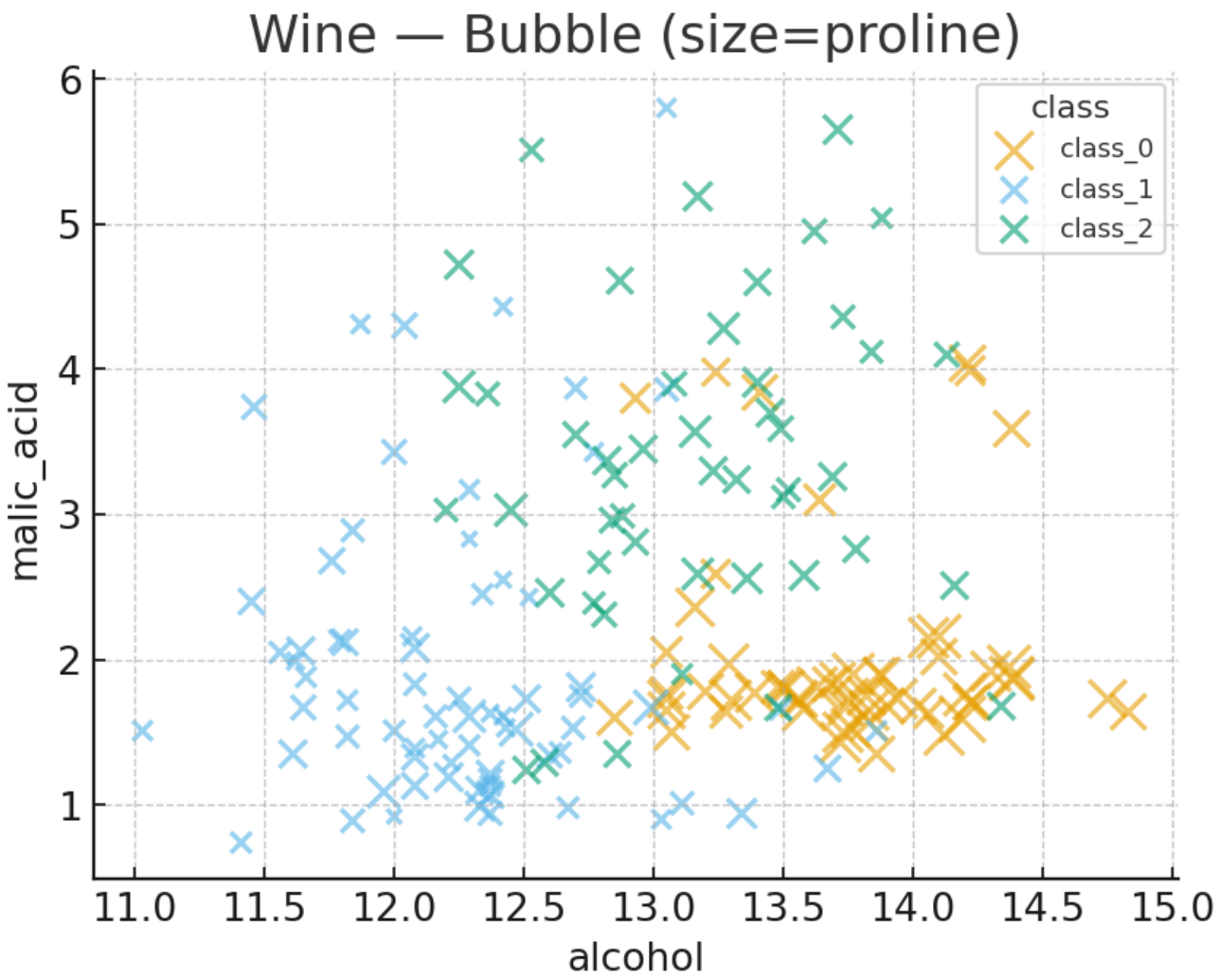


Bubble — Definition & When to Use

Definition. Scatter with marker *size* encoding a third numeric variable; optional *hue* for category.
Use when. Show X–Y relation while comparing magnitude (size) across points.
How to read. Position encodes (x, y); size encodes a third value; color encodes group.
Pitfalls. Human area perception is nonlinear; include legend and avoid huge size ranges.

Bubble — Code & Executed Example

```
sns.scatterplot(data=wine, x="alcohol", y="malic_acid",
                size="proline", hue="class", alpha=0.6)
```

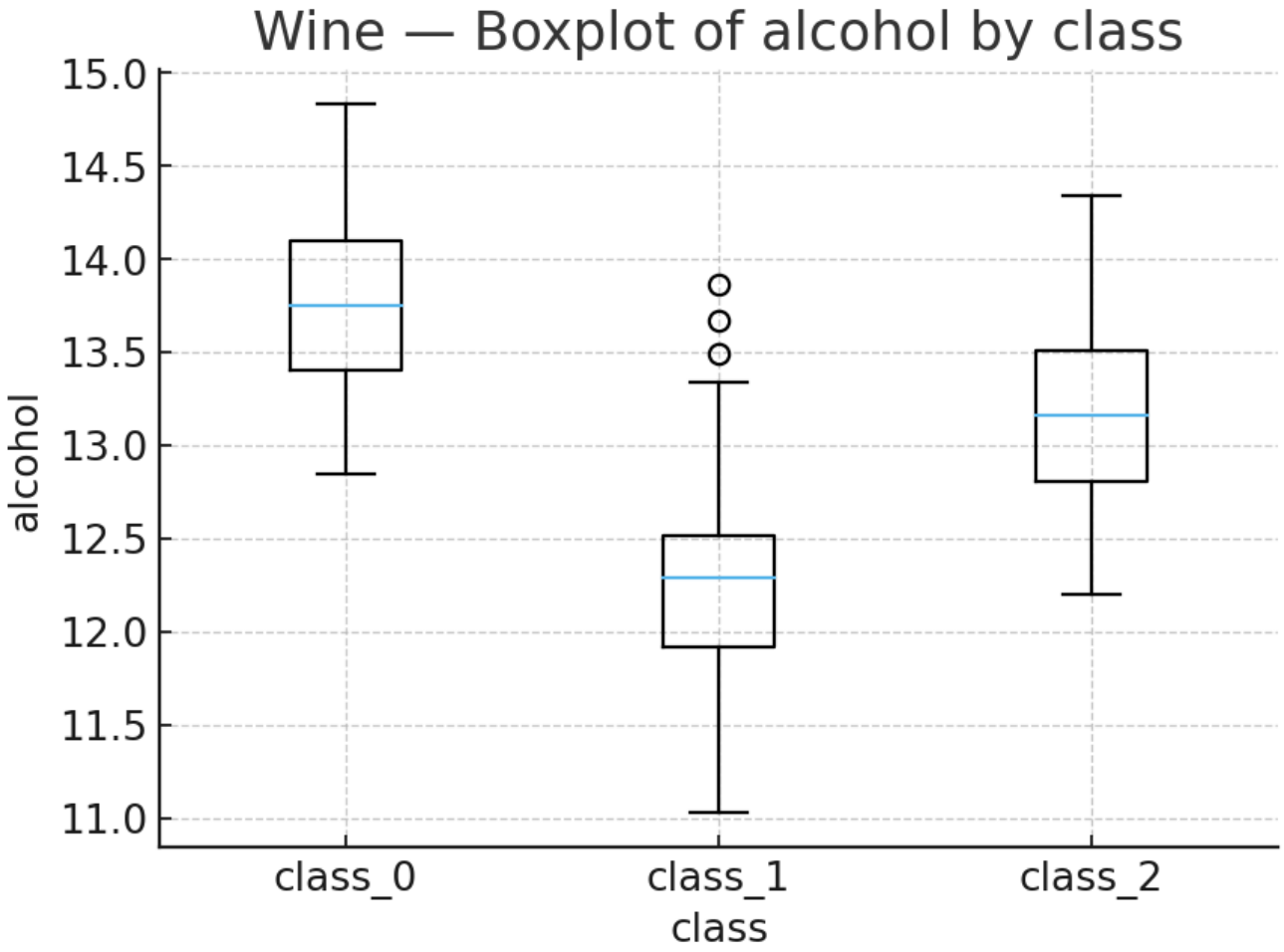


Box — Definition & When to Use

Definition. Median line; box = IQR; whiskers (e.g., 1.5 IQR); points beyond = outliers.
Use when. Robust group comparison of location/spread.
How to read. Median shifts; IQR width; outlier count.
Pitfalls. Hides multimodality; document whisker rule.

Box — Code & Executed Example

```
sns.boxplot(data=wine, x="class", y="alcohol",
            showfliers=True, whis=1.5)
```

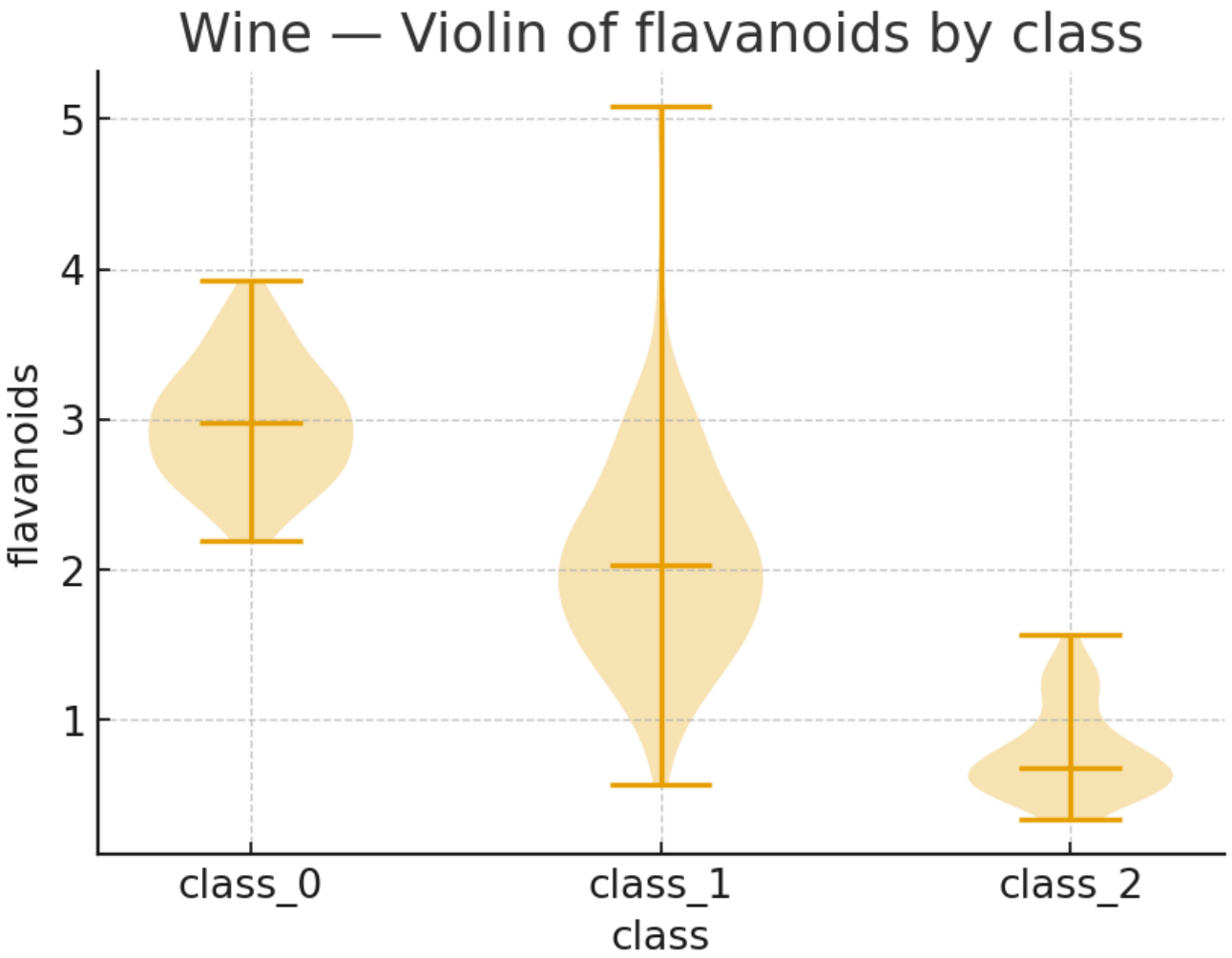


Violin — Definition & When to Use

Definition. KDE envelope mirrored around the center; inner marks for quartiles or median.
Use when. Displaying shape (skew, tails, multimodality) by group.
How to read. Width $\propto$ density; compare widths across y.
Pitfalls. Bandwidth choice matters; avoid over-reading bumps at small n.

Violin — Code & Executed Example

```
sns.violinplot(data=wine, x="class", y="flavanoids",
                inner="quartile", cut=0, bw_adjust=1.0)
```



Boxen — Definition & When to Use

Definition. Letter-value boxes: deeper, symmetric quantile boxes for large n.
Use when. Emphasizing tails with many observations.
How to read. Stacked boxes show deeper quantiles; median line central.
Pitfalls. Requires sufficient n; choose `k_depth` appropriately.

Boxen — Code Example

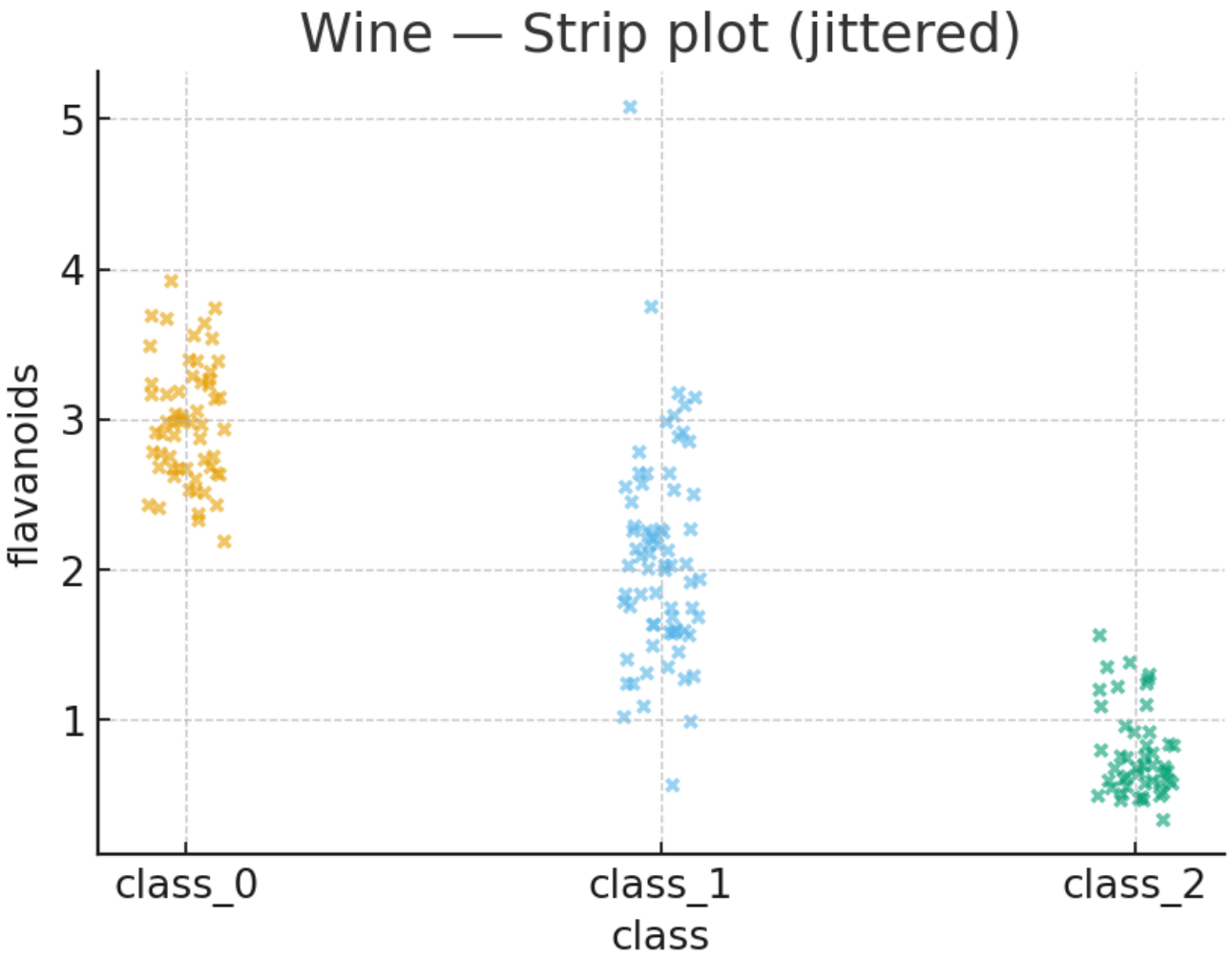
```
sns.boxenplot(data=wine, x="class", y="flavanoids",
              k_depth="trustworthy", width_method="linear",
              showfliers=True, outlier_prop=0.01)
```

Strip/Swarm — Definition & When to Use

Definition. 1D scatter per category (Strip: jittered; Swarm: non-overlapping packing).
Use when. Show raw observations alongside summaries.
How to read. Clumps/gaps reveal microstructure masked by aggregates.
Pitfalls. Overplotting at large n; downsample or combine with summaries.

Strip — Code & Executed Example

```
sns.stripplot(data=wine, x="class", y="flavanoids",
              jitter=0.25, alpha=0.6, size=3)
```

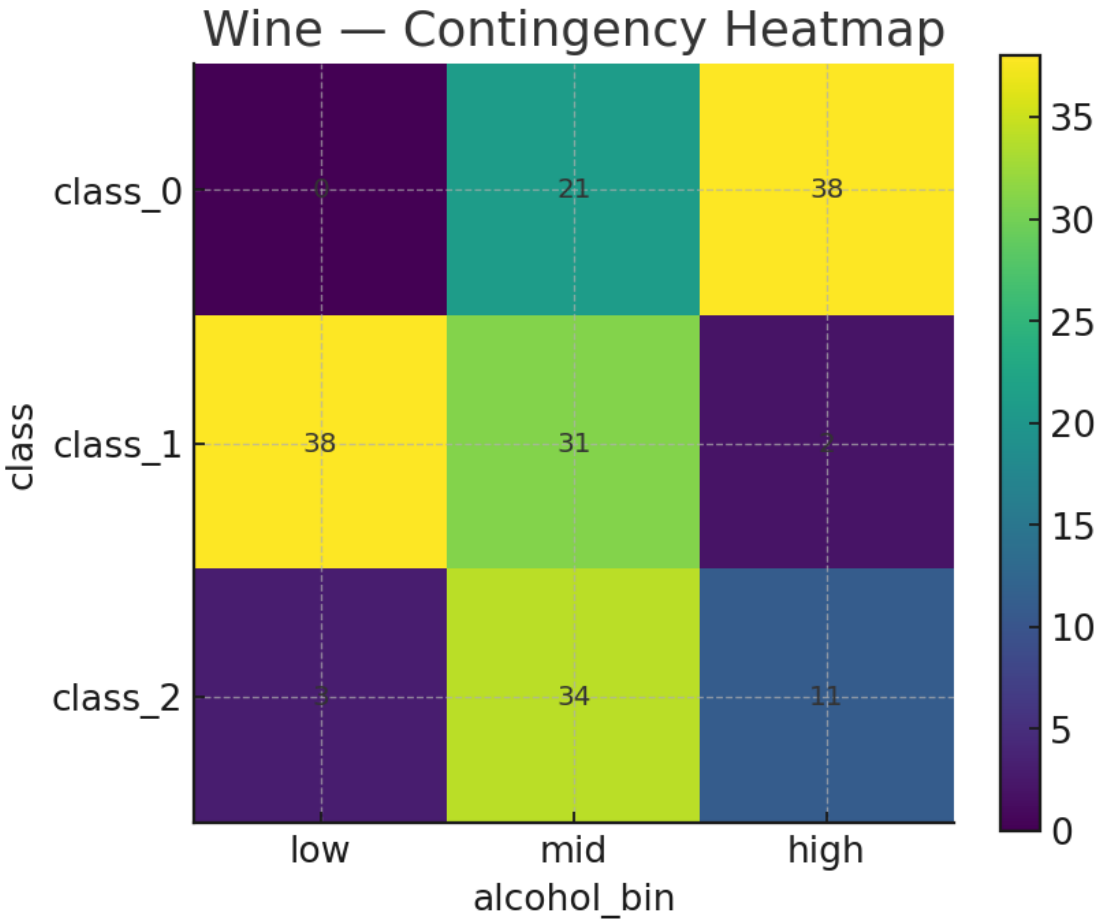


Contingency Heatmap — Definition & When to Use

Definition. Heatmap of a cross-tabulation; color encodes counts or rates.
Use when. Assess association/imbalance for two categorical variables.
How to read. Row/column patterns; normalize for rate comparison.
Pitfalls. Unequal group sizes: prefer percentages over raw counts.

Contingency Heatmap — Code & Executed Example

```
wine["alcohol_bin"] = pd.cut(wine["alcohol"], 3, labels=["low","
    mid","high"])
ct = pd.crosstab(wine["class"], wine["alcohol_bin"])
sns.heatmap(ct, annot=True, fmt="d")
```

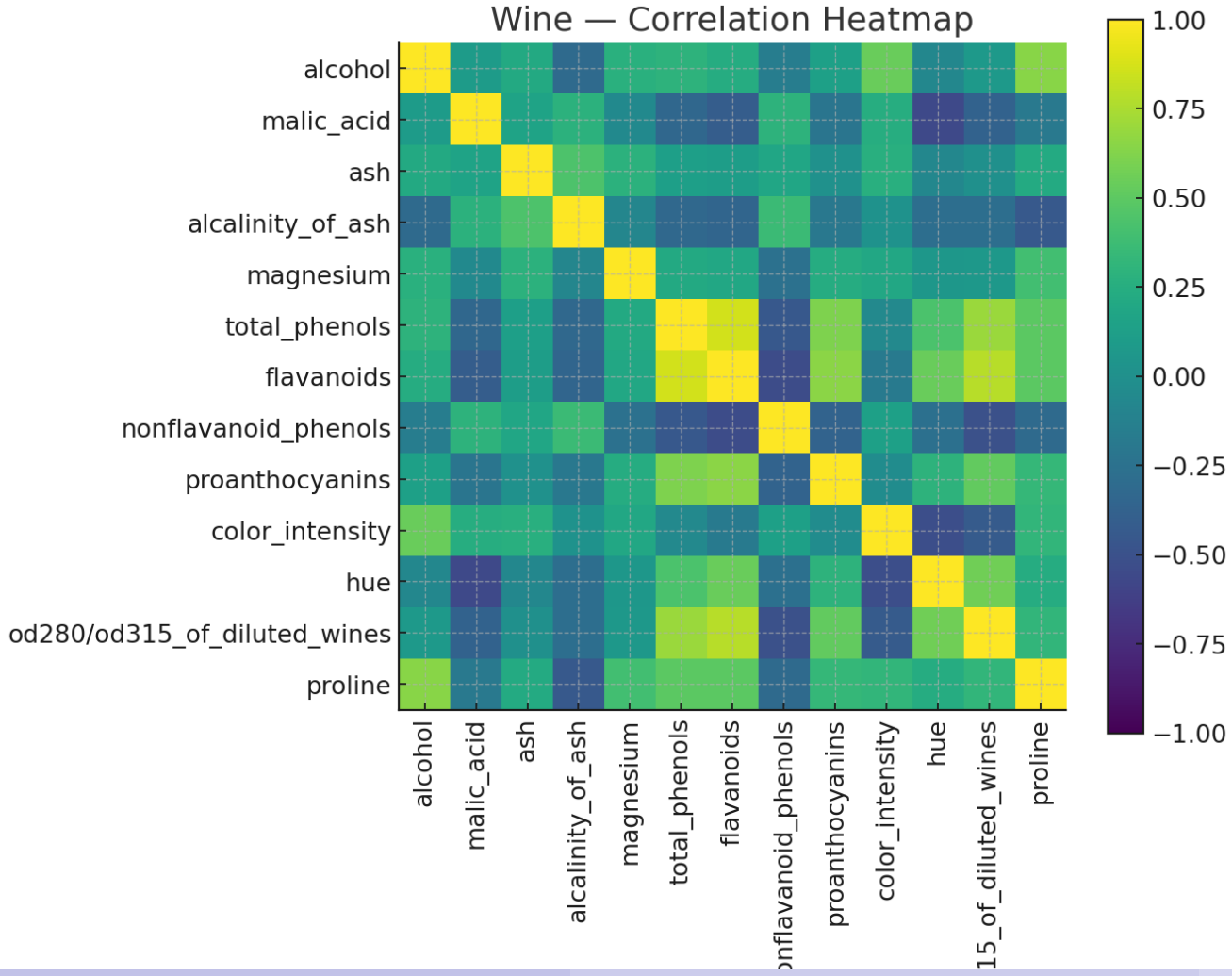


Correlation Heatmap — Definition & When to Use

Definition. Matrix of Pearson r between numeric features; color encodes strength/sign.
Use when. Scan linear associations, multicollinearity.
How to read. Blocks/stripes $\Rightarrow$ feature groups.
Pitfalls. Correlation $\neq$ causation; outliers inflate $|r|$.

Correlation Heatmap — Code & Executed Example

```
num = wine.drop(columns=["target", "class"])
corr = num.corr()
sns.heatmap(corr, vmin=-1, vmax=1, center=0, cmap="vlag")
```

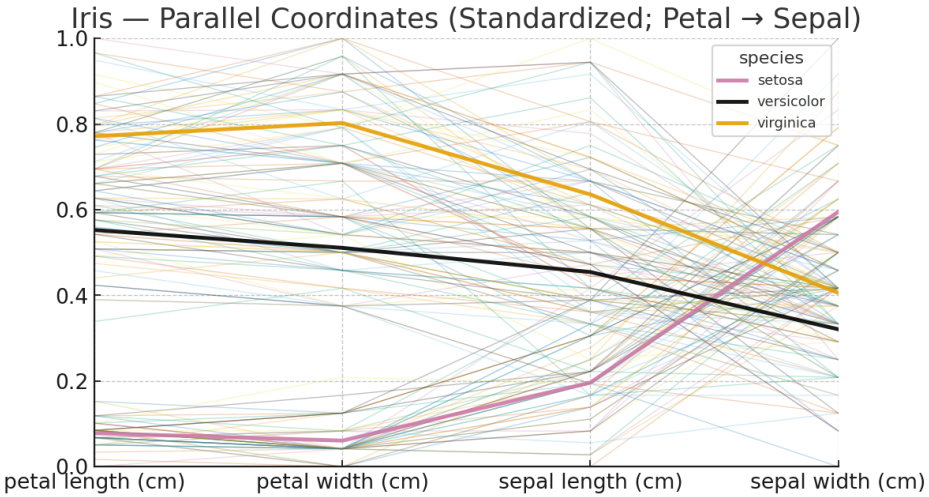


Parallel Coordinates — Definition & When to Use

Definition. Each sample is a polyline across feature axes (features are aligned on parallel vertical axes).
Use when. Compare *multivariate profiles* by class; spot axes with strong separation or crossings.
How to read. Separation along an axis $\Rightarrow$ discriminative feature; frequent crossings $\Rightarrow$ interaction/trade-offs across features.
Pitfalls. Clutter at large n; *standardize* features, use transparency, downsample or aggregate.

Parallel Coordinates — Executed Example (Iris, standardized, Petal → Sepal)

```
# Standardize in this exact order
X = iris[features].to_numpy()
X = StandardScaler().fit_transform(X)
df_scaled = pd.DataFrame(X, columns=features)
df_scaled["species"] = iris["species"]
plt.figure(figsize=(7.6, 4.4))
parallel_coordinates(df_scaled, "species", features, lw=0.9, alpha=0.45)
plt.tight_layout()
plt.show()
```



Analysis. *Setosa* has smaller **Petals**, but its **Sepal** tends to be wider.

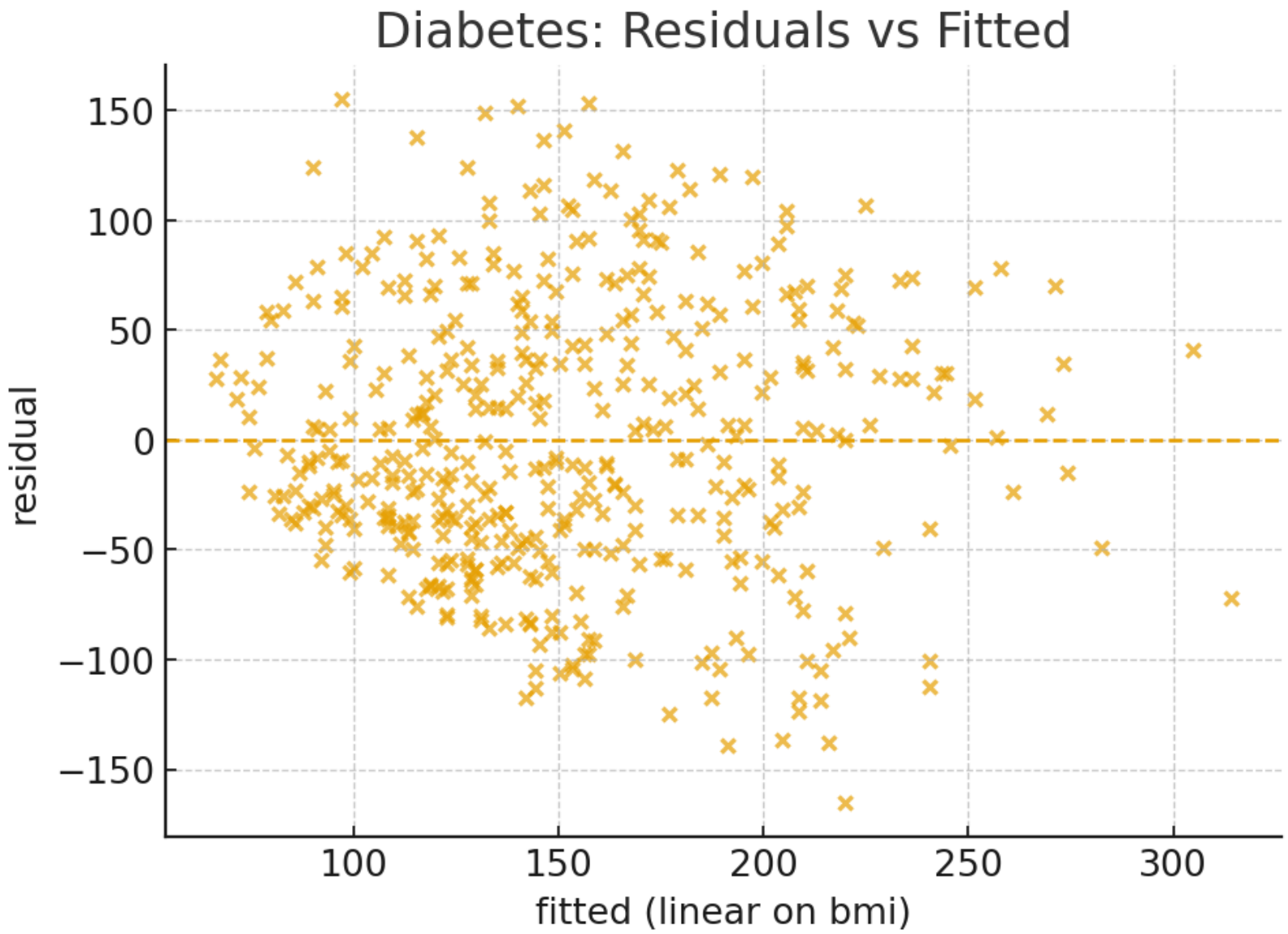
Residuals vs Fitted — Definition & When to Use

Definition. Plot $(\hat{y}_i,\ y_i - \hat{y}_i)$ to assess model assumptions.
Use when. Linear diagnostics (nonlinearity, heteroscedasticity, outliers).
How to read. Random band around 0 is good; trends/fans suggest issues.
Pitfalls. Check leverage/influence separately (Cook's distance).

Residuals vs Fitted — Code (Seaborn)

```
# Basic linear fit: target ~ bmi
x = diabetes["bmi"].to_numpy()
y = diabetes["target"].to_numpy()
slope, intercept = np.polyfit(x, y, 1)
fitted = slope * x + intercept
residuals = y - fitted

# Seaborn plot: residuals vs fitted
sns.set_theme()
ax = sns.scatterplot(x=fitted, y=residuals, alpha=0.7)
ax.axhline(0, linestyle="--", linewidth=1.2)
ax.set(
    xlabel="fitted (linear on bmi)",
    ylabel="residual",
    title="Diabetes: Residuals vs Fitted"
)
plt.tight_layout()
plt.show()

# Alternative (classic residual plot: predictor on x-axis):
# sns.residplot(data=diabetes, x="bmi", y="target")
```

Residuals vs Fitted — Executed Example